IS1212 : Probability and Statistics

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Organizing and Presenting Data

Organizing & Presenting Data

- Raw data provide little information.
- Need to convert the raw data into useful information.
  - MFMFMFMMMMMFMMFMFMMMFMFMMMFMMF
  - FFFFFFFFFFMMMMMMMMMMMMMMMMMMMMM
  - 10F and 20M
- Comparing data is easier when it is clear and specific.
- Organizing data makes easy to read and understand.
- Charts, plots, graphs and tables are ways to organize/present data
- Graphs and charts helps to get the attention of the audience.

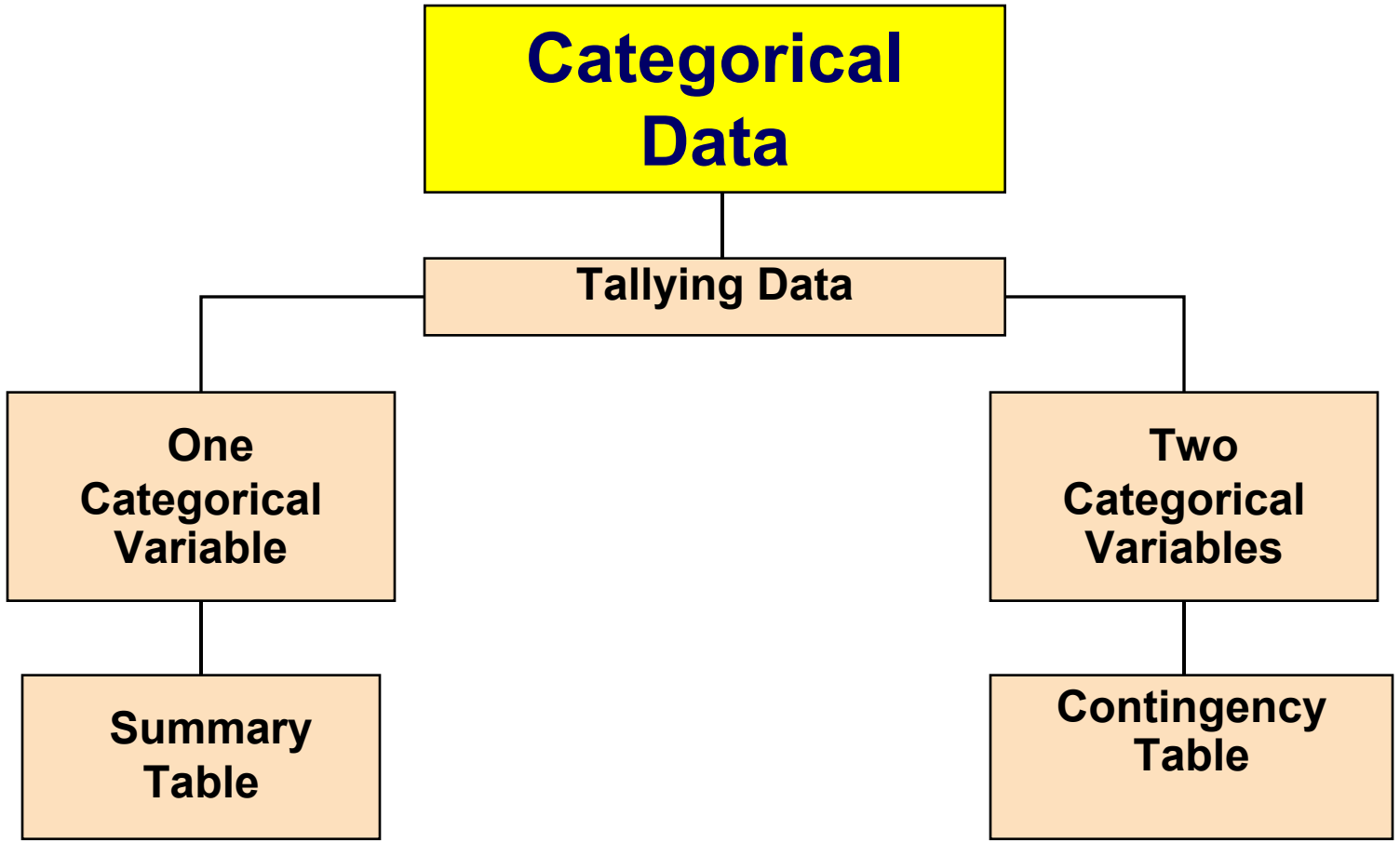
Presenting Data

- **Tabular format**
  - Presenting information in a "Simple" summarized form
    - One way Tables (Summary Table/Frequency table)
    - Cross Contingency Tables (Multidimensional tables)
      - Two way tables
      - Three way tables

- **Pictorial format**
  - Diagrams, charts or graphs.
  - Does not supply any more information than the tabular form
  - Illustrates the important facts more clearly

Ways to Present Categorical Data

Ways to organize Categorical Data



Tabular form

- One-Way Table

District	Number of COVID19 Patients	Percentage
Colombo	25	43.1
Gampaha	16	27.6
Kaluthara	14	24.1
Kandy	2	3.4
Mathara	1	1.7
Galle	0	0.0
Anuradhapura	0	0.0
Polonnaruwa	0	0.0
Total	**58**	**100**

Age	Frequency	Percentage
41 - 45	8	18.6
46 - 50	10	23.3
51 - 55	10	23.3
56 - 60	7	16.3
61 - 65	6	14.0
66 - 70	2	4.7
Total	**43**	**100**

Tabular form…

- Two-way Table
  - Frequency Table
  - Overall Percentages
    - $\frac{Count}{Overall\ Total} \times 100\%$
  - Row Percentages
    - $\frac{Count}{Row\ Total} \times 100\%$
  - Column Percentages
    - $\frac{Count}{Column\ Total} \times 100\%$

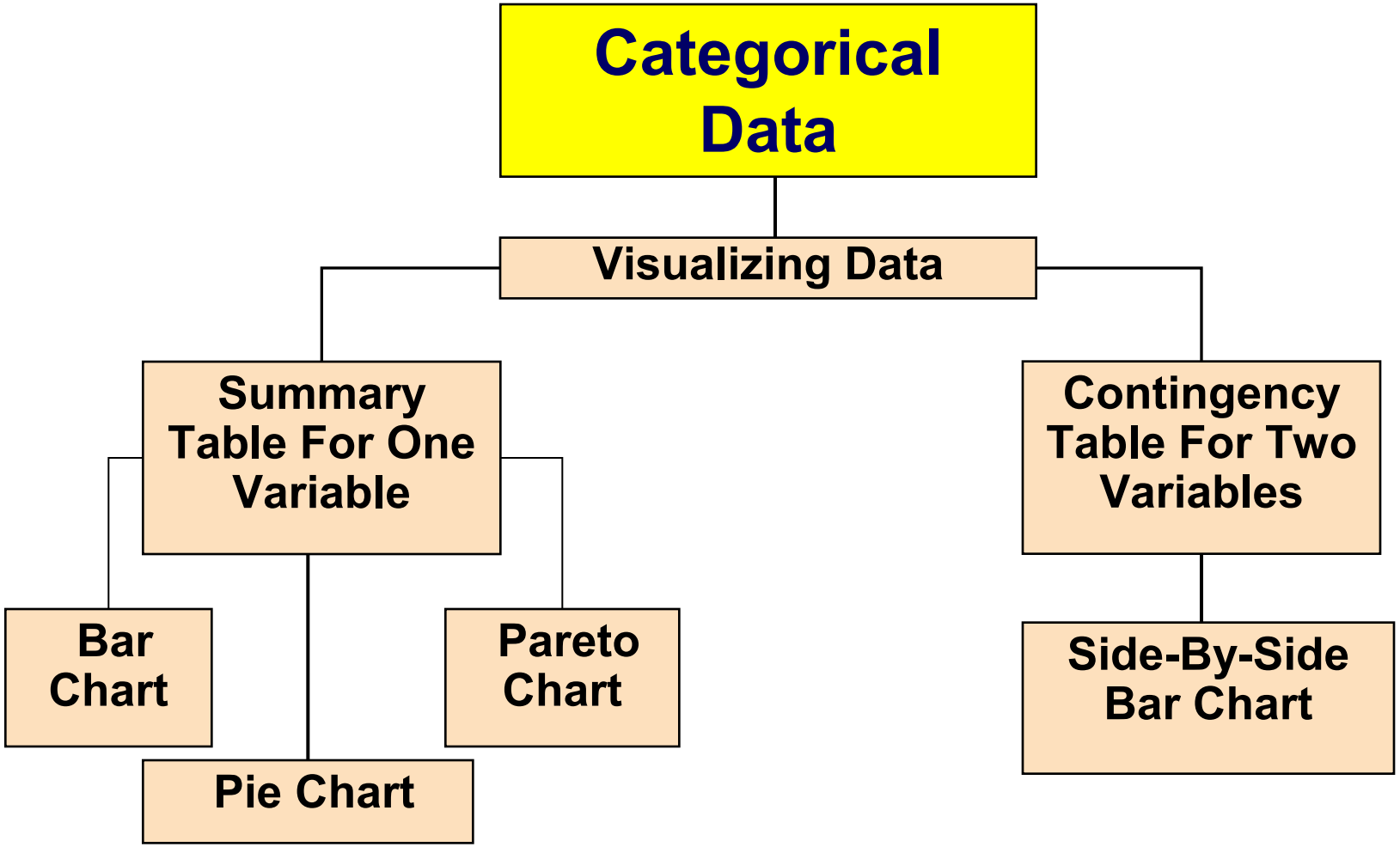
		District			
		Colombo	Gampaha	Kaluthara	Total
Gender	Male	15	10	9	**34**
	Female	10	6	5	**21**
	Total	**25**	**16**	**14**	**55**

		District			
		Colombo	Gampaha	Kaluthara	Total
Gender	Male	27.3	18.2	16.4	**61.8**
	Female	18.2	10.9	9.1	**38.2**
	Total	**45.5**	**29.1**	**25.5**	**100**

		District			
		Colombo	Gampaha	Kaluthara	Total
Gender	Male	44.1	29.4	26.5	**100.0**
	Female	47.6	28.6	23.8	**100.0**

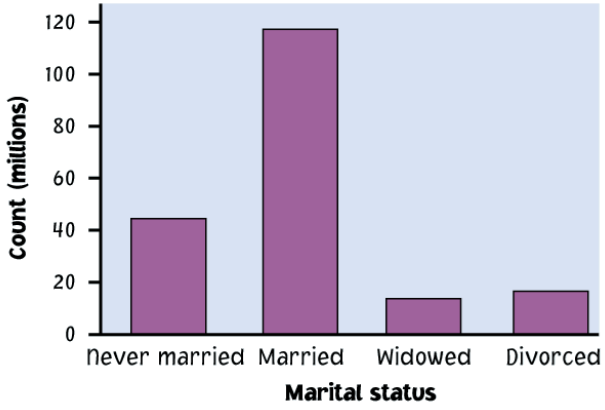
		District		
		Colombo	Gampaha	Kaluthara
Gender	Male	60.0	62.5	64.3
	Female	40.0	37.5	35.7
	Total	**100.0**	**100.0**	**100.0**

Ways to Present Categorical Data



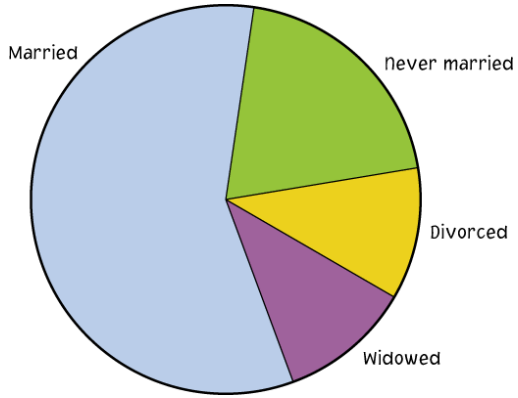
Ways to Present Categorical Data

- Categories can be ordered any way that we want (alphabetical order, increasing order, by year, etc.)



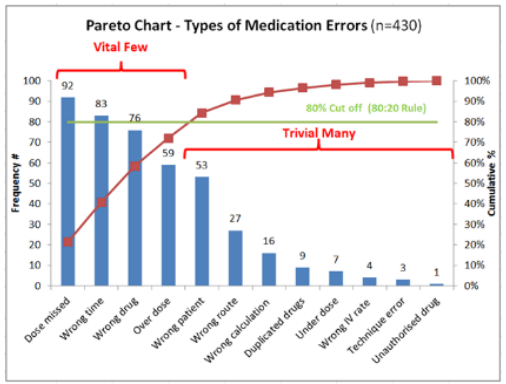
- Bar/Column Charts

  Each category is represented by a bar/Column



- Pie Charts

  Slices represent the parts of one whole



- Pareto Chart

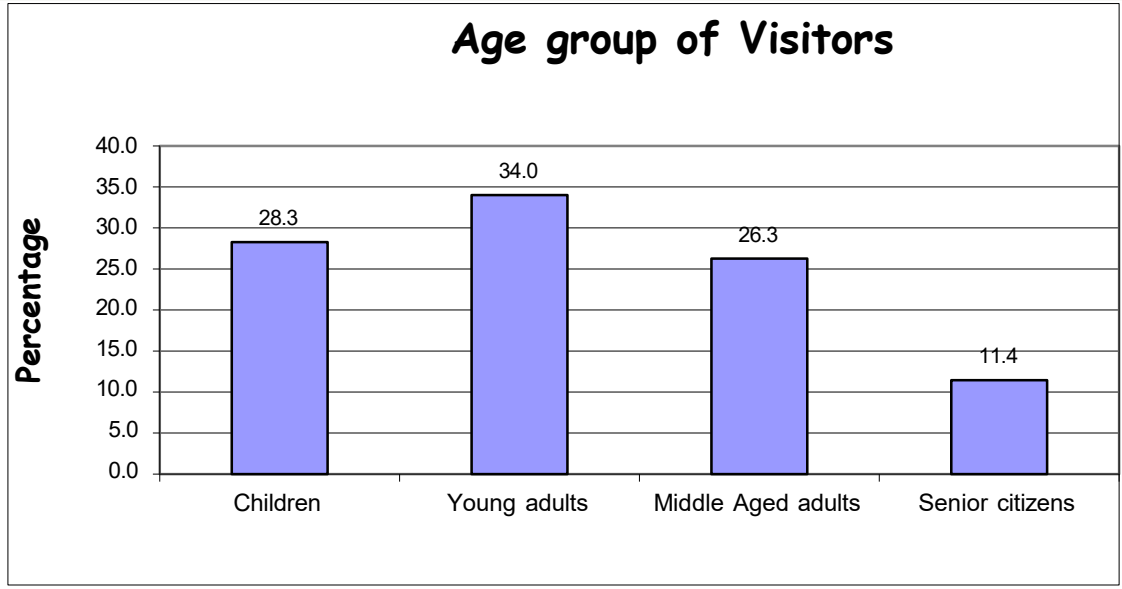
  Each category is represented by a bar in decreasing order and with the cumulative frequency line

Ways to Present Categorical Data…

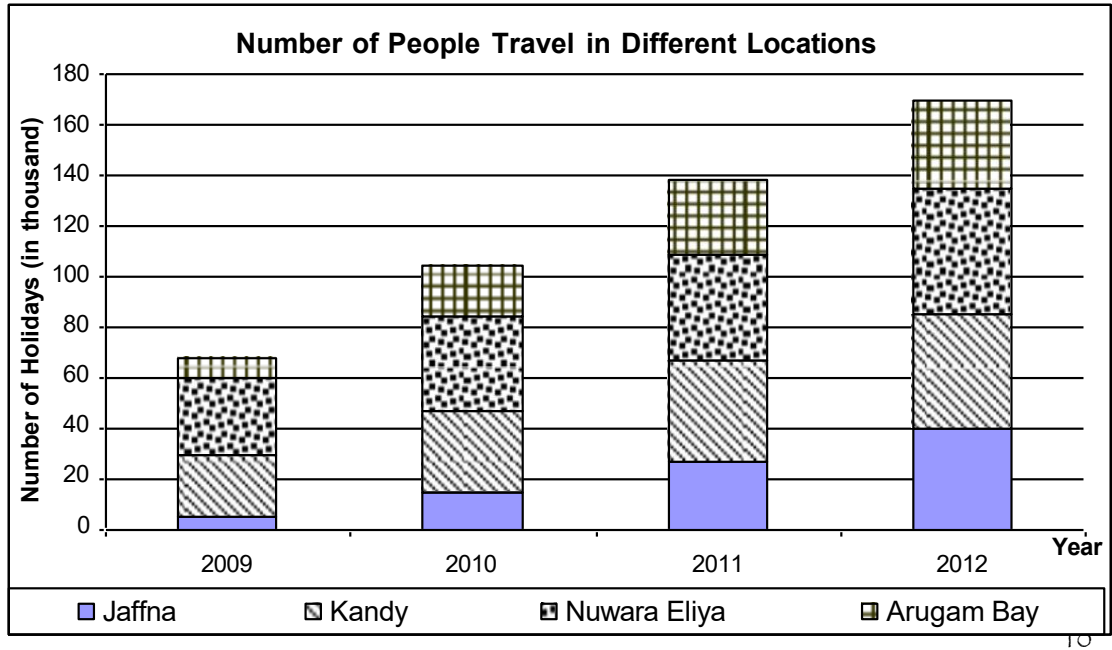
- Bar chart
  - Use parallel bars (rectangles) on horizontal axis.
  - Height of a bar equals to the count/percentage.
  - Types of bar charts
    - Simple bar charts
    - Component bar charts
    - Percentage component bar chart
    - Multiple bar charts

Ways to Present Categorical Data…

- **Simple Bar Chart**
  - Series of bars
  - Height of each bar is the frequency/percentage



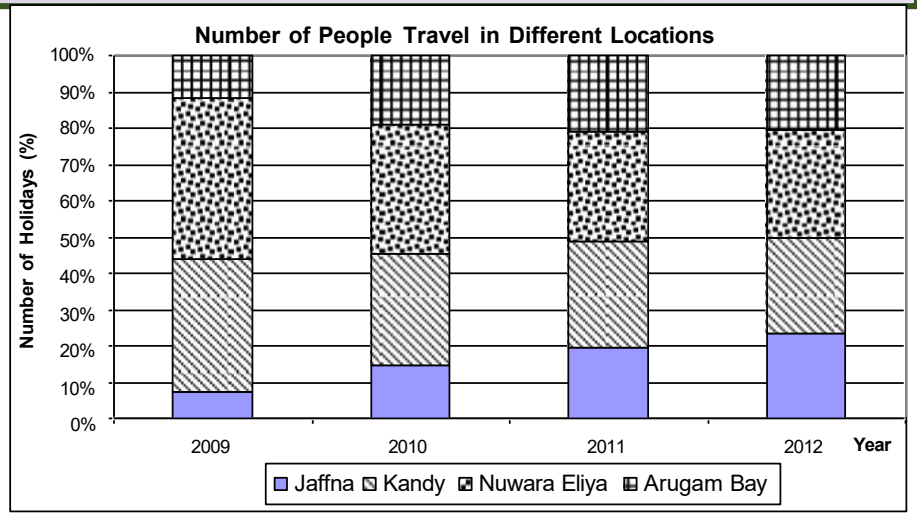
- **Component Bar Chart**
  - Bars are sub divided into components.
  - To see the changes in totals and indication of size of the component figures.

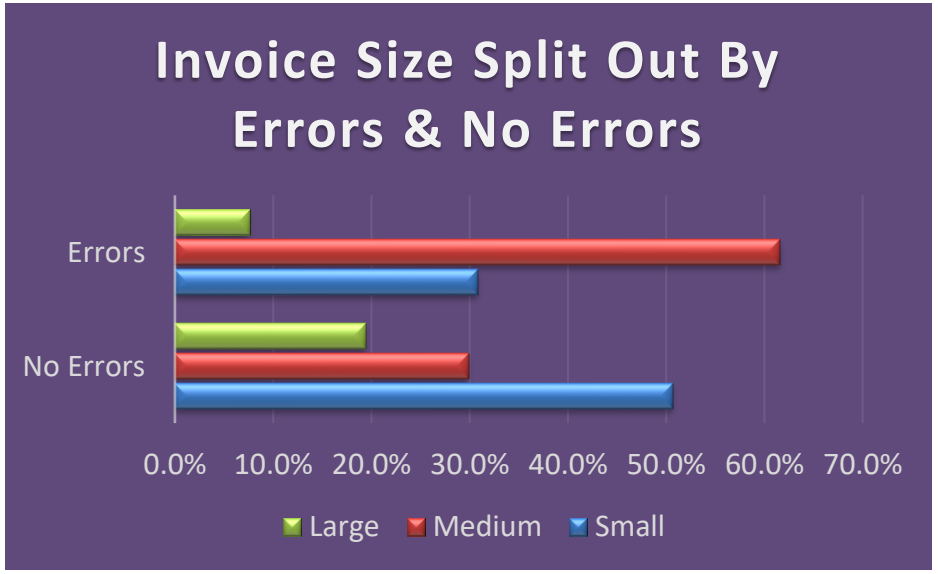


Ways to Present Categorical Data…

- **Percentage Component Bar Chart**
  - When changes in relative size of component figures are required

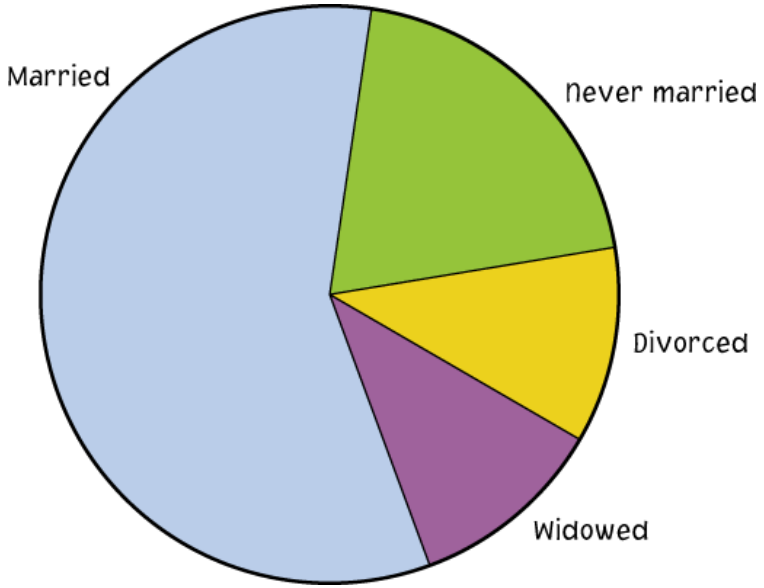
- **Multiple Bar Chart**
  - Component figures (Count/ Percentage) shown as separate bars adjoining to each other.
  - Height of each bar is actual value of the component figure.
  - To see changes in the actual values of the component figures only are required and overall total is not important.





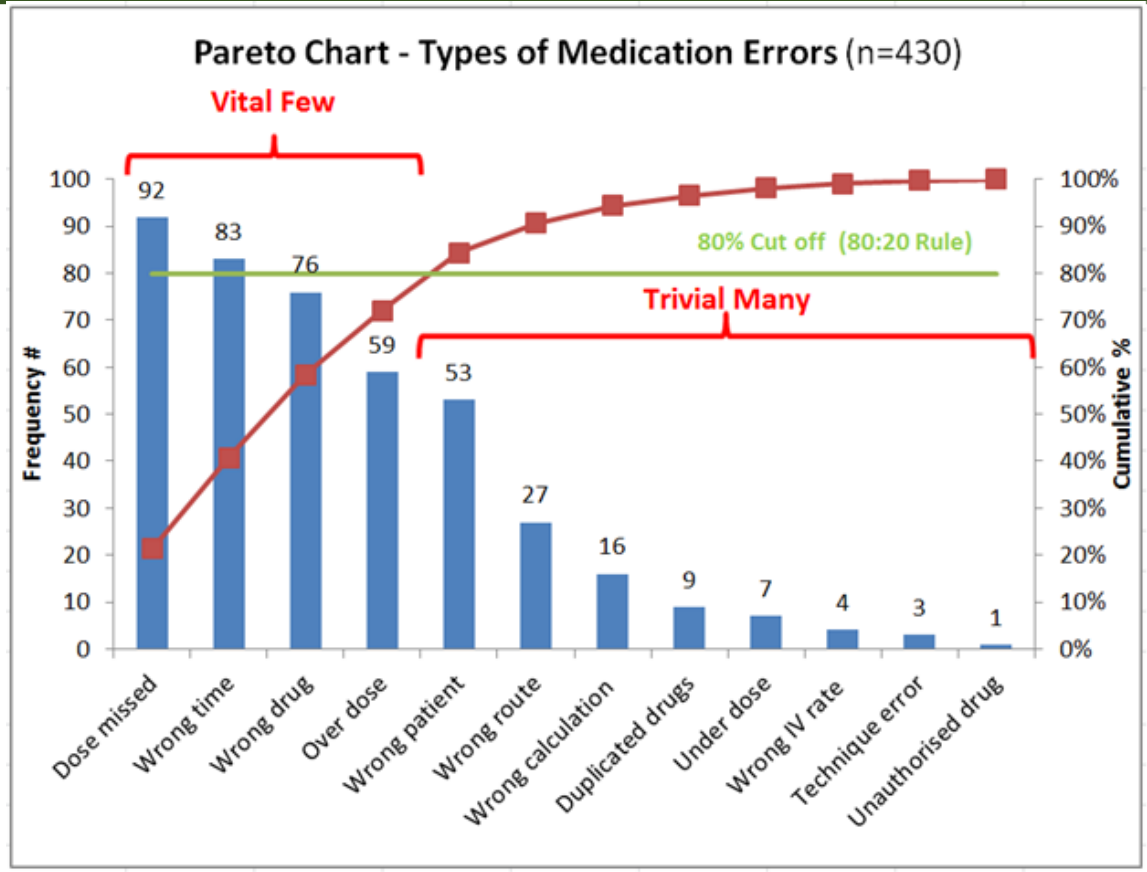
Ways to Present Categorical Data…

- **Pie Chart**
  - Component parts
  - Represents by a circle
  - Expressed as percentages
  - Sizes of segments of the circle are proportional to the percentages
  - Not suitable, if number of categories are large (say 6)



Ways to Present Categorical Data…

- **Pareto Chart**
  - Bars are in decreasing order of frequency from left to right
  - 80/20 Rule **(Pareto principle/law of the vital few & trivial many)**
    - Roughly 80% of the effects come from 20% of the causes.



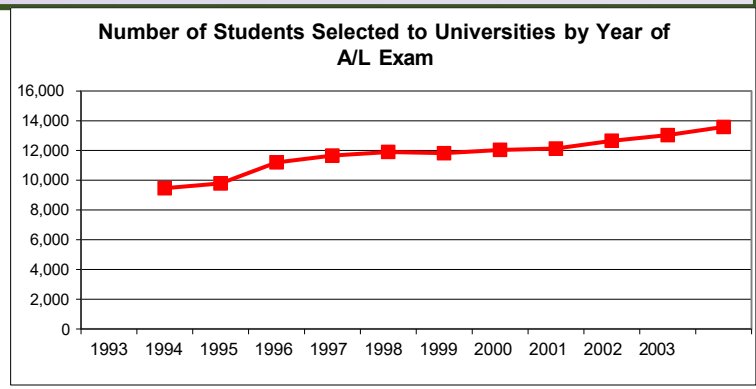
Ways to Present Numerical Data

Ways to Present Numerical Data

- Time plots and Line chart
  - When there is a meaningful sequence, like time or in order.
  - Lines connected points helps to find out changes over time.
- Histograms
  - Summary graph for a single numerical variable (Frequency Distribution).
  - Useful to understand the pattern of variability in the data.
- Stem-and-Leaf Plots
  - Useful to understand the pattern of variability in the data.
- Box-Plot
  - Represents the five number summary of the data
- Scatter Diagram
  - Represents the relationship between two variables

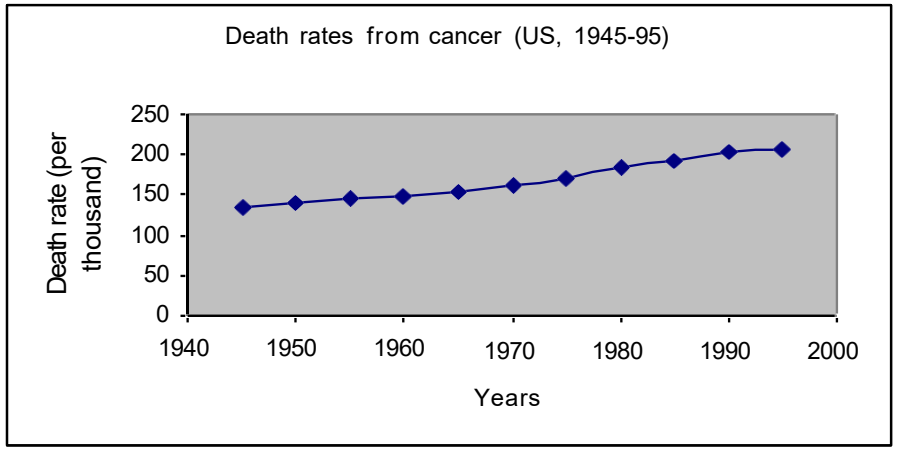
Ways to Present Numerical Data…

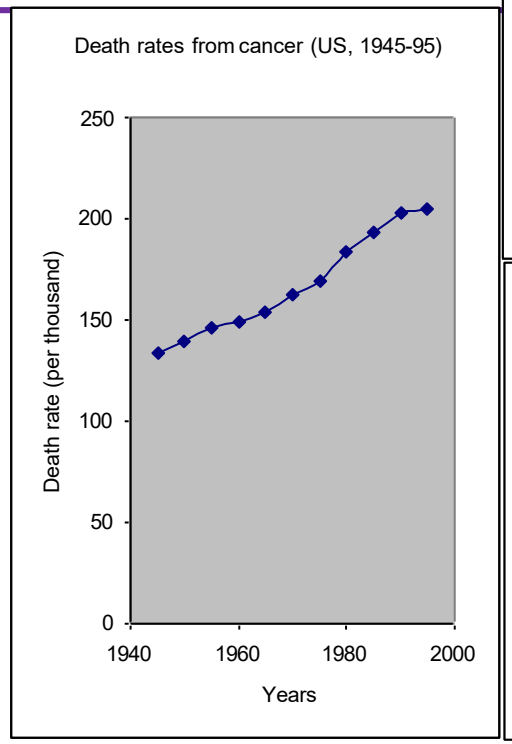
- **Time Plot**
  - Always "Time" on the horizontal axis
  - Describe by looking at overall pattern
  - In a time series:
    - **trend** : a rise or fall over time, despite small irregularities.
    - **seasonal variation** : pattern repeats at regular intervals.
- **Line Chart**
  - To emphasize the changes in some variable occurring in time or any order.
  - It is obtained by plotting as points and joining consecutive points

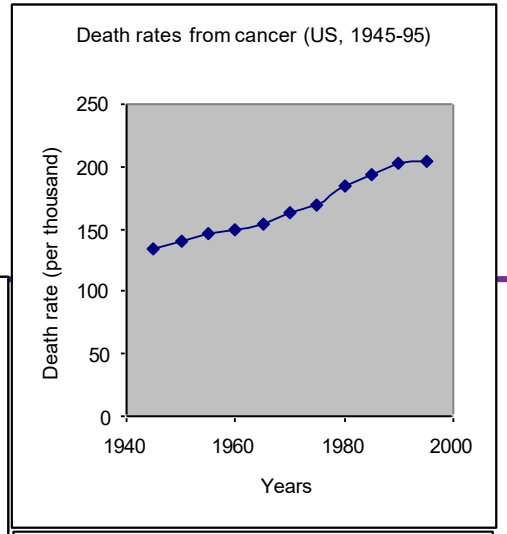


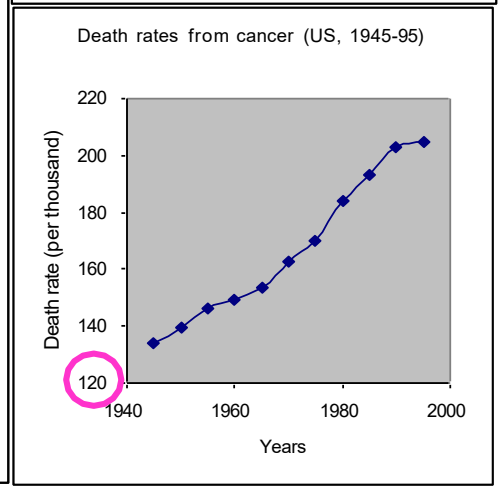
Scales Matter

- Stretching the axes and choose of scales can give a different impression.









Frequency Distribution

- The easiest method of organizing numerical data is a frequency distribution

- **Basic Steps of Constructing a Frequency Distribution (one method)**
  - Step 01 : Calculate the range (R)
  - Step 02: Specify the number of classes (k)
    - There is no accepted rule.
    - Between 5 and 15 classes are recommended.
    - No overlapping.
  - Step 03: Determine the class width (W), W=R/k Round up!
  - Step 04: Determine class boundaries
  - Step 05: Count the frequencies

Example

- The frequency table is as follows:

Class Interval	Frequency	Rel. Freq.	Percentage
$18 \leq x < 20$	3	.30	30%
$20 \leq x < 22$	2	.20	20%
$22 \leq x < 24$	4	.40	40%
$24 \leq x < 26$	1	.10	10%

- Sum of all the relative frequency = 1.00

More About Frequency Distribution

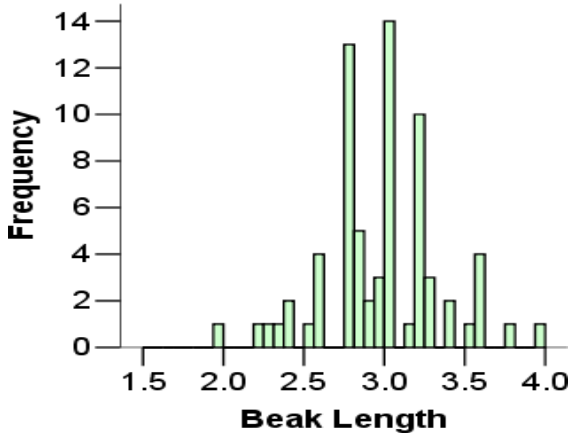
- Class mark
  - Class mark is the mid-point of the class interval.
  - It is obtained by averaging the lower and upper class boundaries.

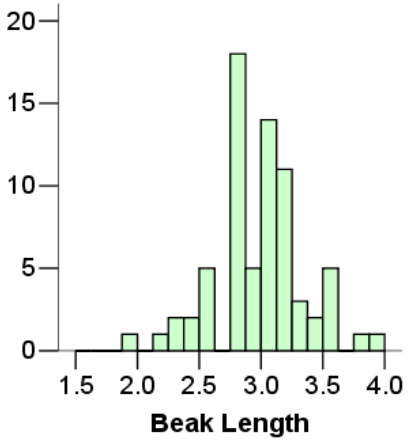
- **Cumulative Frequency**
  - Total frequency up to and including the upper boundary of a given class interval

Histogram

- Graphical representation of a frequency distribution.
- It consist of a set of rectangles (bars),
  - On the X-axis centered at the class mark
  - Length equal to the class width.
  - Area is proportional to the class frequency.
    - When class widths are equal, the heights of the rectangles are equals to the class frequency.
    - When class widths are unequal, the heights of the rectangles are proportional to the frequencies.
    - Frequency density is used instead of frequency, where; frequency density = Frequency/Class width

Guide to Create a Histogram…

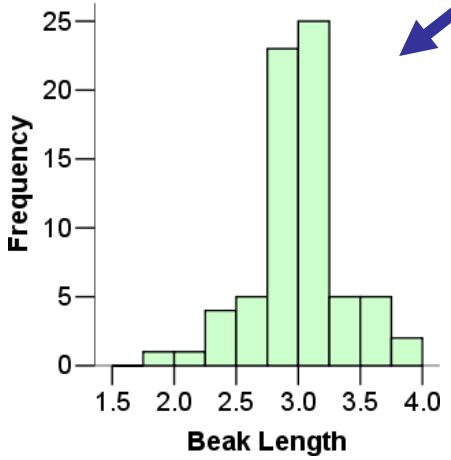


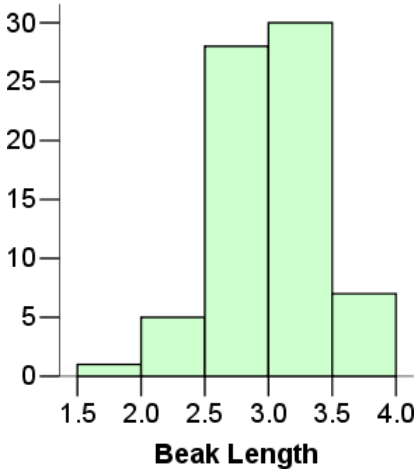


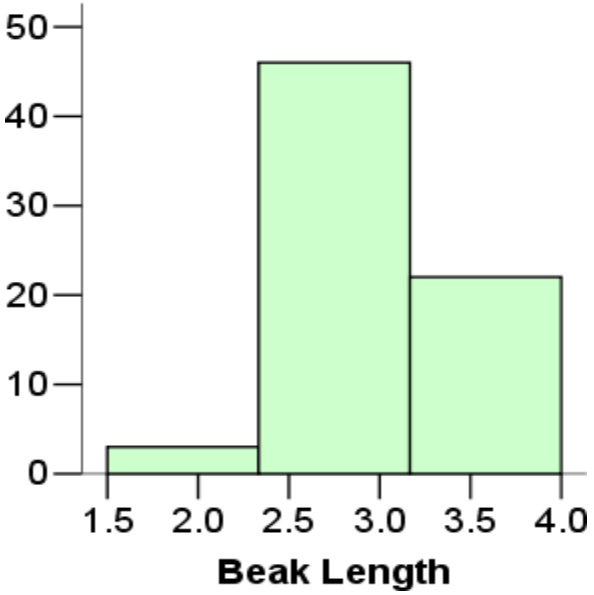
Not summarized enough

inappropriately summarized

Summarized Well







Frequency Polygon
Cumulative Frequency Polygon (Ogive)

- **Frequency Polygon**
  - Plot of frequency density against the class mid point.
  - Join the points by straight line segments.
  - Provides a useful way of comparing two or more data sets.

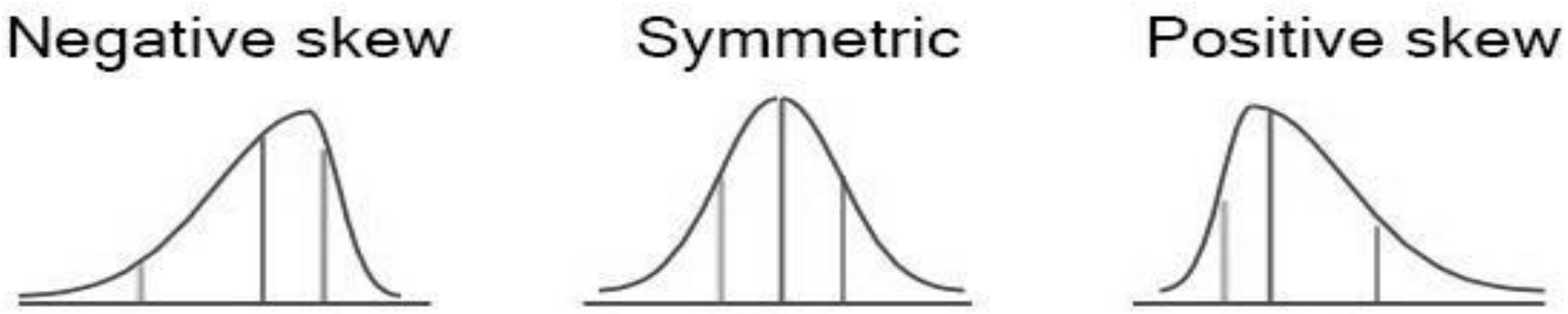
- **Cumulative Frequency Polygon (Ogive)**
  - Y- axis : cumulative frequencies
  - X- axis : upper class boundaries

Frequency Curve

- When the number of intervals gets large, the frequency polygon will consist of a large number of line segments, and the frequency polygon approaches a smooth curve.
- Smoothed frequency polygon.
- Useful to have some idea about the shape of the freq. distribution.
- **Two Aspects of the Shape of a Frequency Curve**
  1. Skewness
  2. Kurtosis

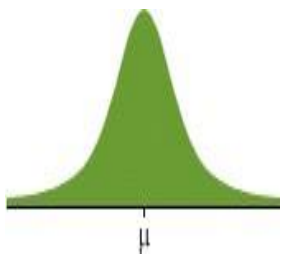
Skewness

- Measure of the lack of symmetry (asymmetry).
- A distribution is symmetric if it looks the same to the left and right of the center point.
- When a frequency curve is asymmetric, one tail of the curve is longer than the other.
  - If it has a long right tail then it is positively skewed.
  - If it has a long left tail then it is negatively skewed.



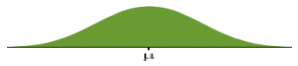
Kurtosis

- Measures whether the data are peaked or flat relative to a normal shape.
- Data sets with high kurtosis;
  - tend to have a distinct peak near the mean,
  - decline rather rapidly,
  - and have heavy tails.



High kurtosis

- Data sets with low kurtosis;
  - tend to have a flat top near the mean,
  - rather than a sharp peak.



Low kurtosis

Stem and Leaf Plot

- Simultaneously organizes the data for analyses and presents the data in a graphical form.
- Presents a frequency distribution and a histogram in one display.
- Gives quick picture of the sample of a distribution
- Work best for small numbers of observations.
- Each numerical value is divided into 2 parts:
  - Leading digit(s) : stem
  - Trailing digit : leaf
- Stems are located along the main axis.
- Leaves are located to represent the distribution.

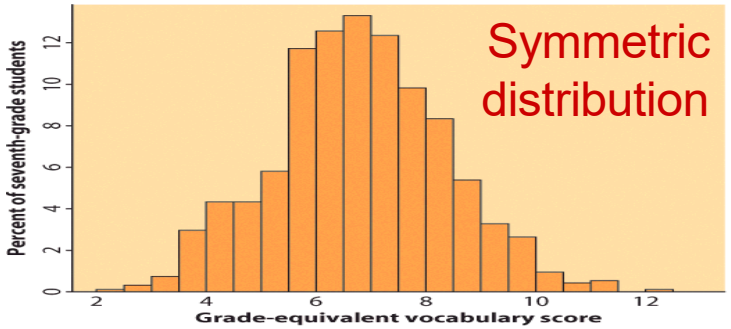
STEM	LEAVES
0	9 9
1	
2	2
3	2 3 9 9
4	2 9
5	2 8
6	
7	0

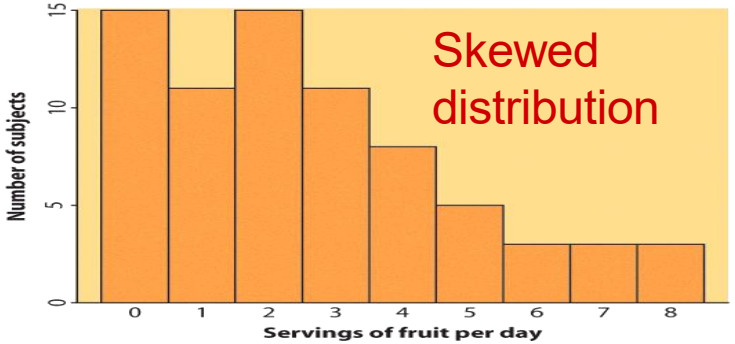
Common Shapes of Distribution

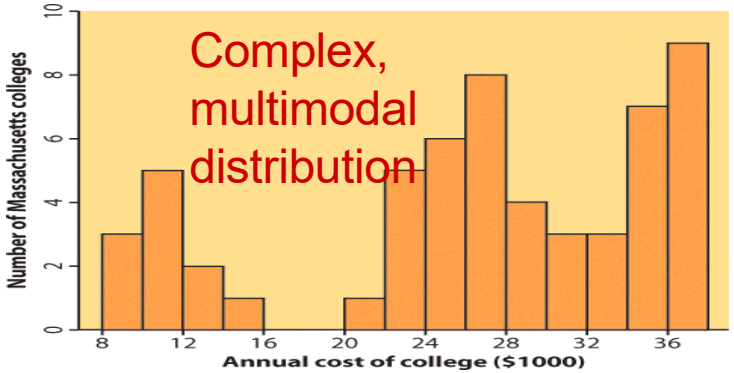
Most Common Distribution Shapes

- Symmetric: the right and left sides of the histogram are approximately same.
- Right Skewed: the right hand side of the histogram extends much farther out than the left side.
- Left Skewed: left hand side of the histogram extends much farther out than the right side.
- Not all distributions have a simple overall shape, especially when there are few observations.

Most Common Distribution Shapes…



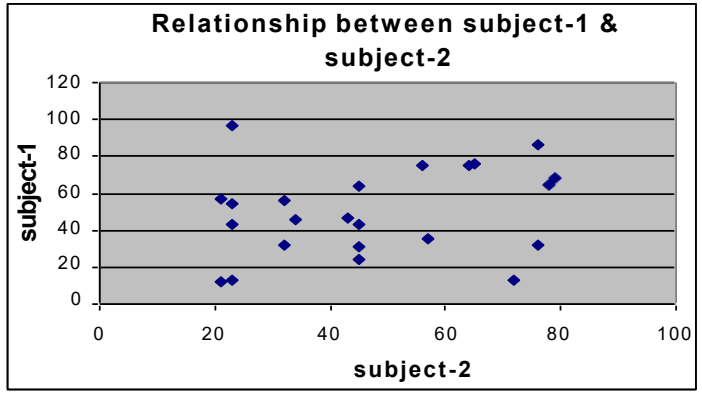


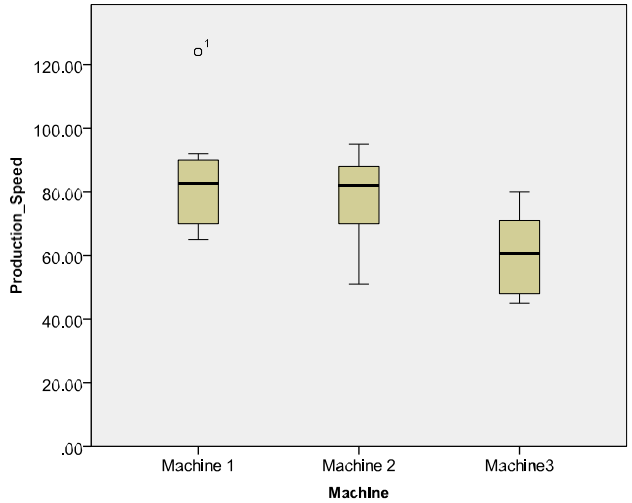


Presenting Relationship Between Two Variables

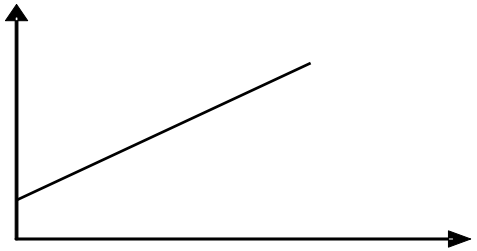
Relationship between Two Variables

- Comparing **shape, centre and spread**.
- The relationship can be viewed graphically;
  - **Scatter Diagram (two continuous variables)**
    - Plot of the ordered pairs of the two continuous variables.
  - **Parallel Box Plot (one continuous variable and the other one categorical variable)**
    - One box plot is displayed for each category

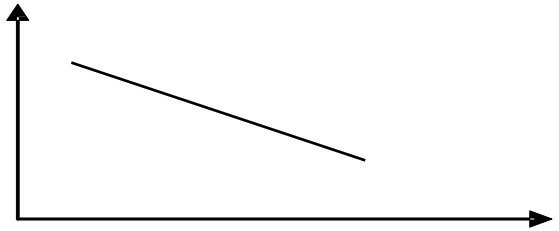




Types of Relationships

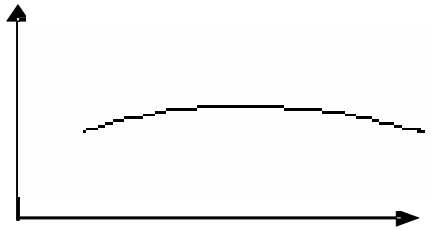


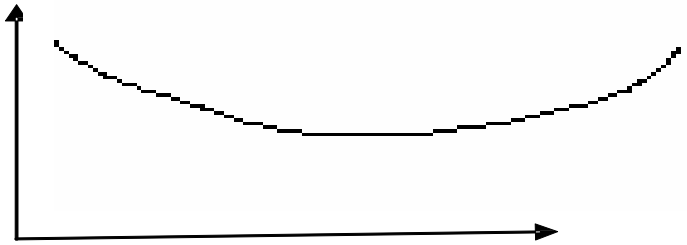
Increasing trend



Decreasing trend

There is a linear relationship

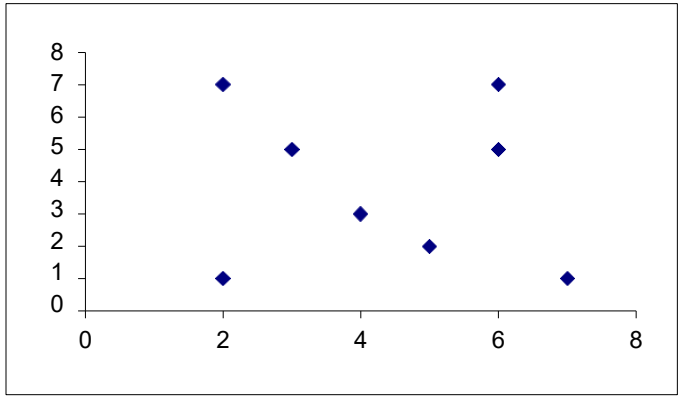




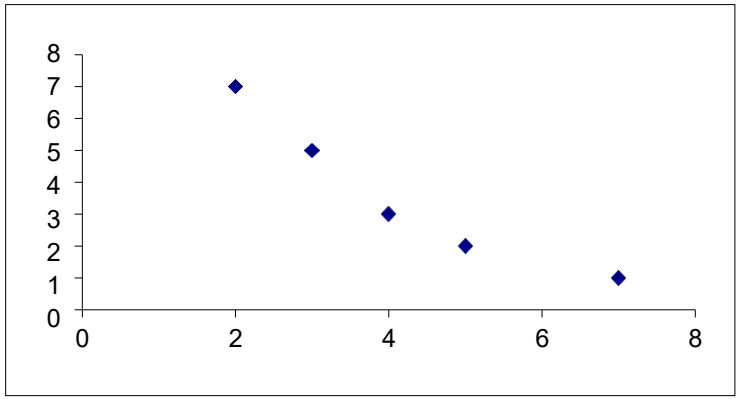
No linear relationship between two variables

How to Interpret a Scatter Plot

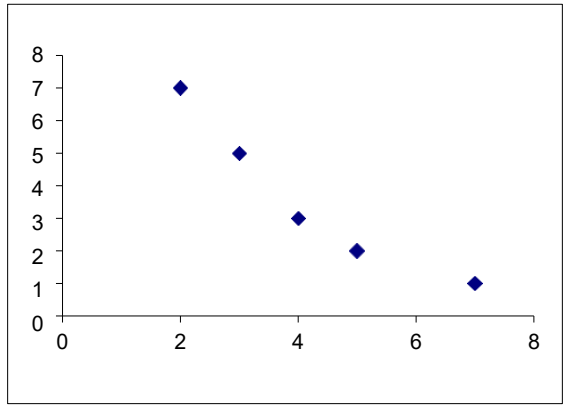
Look if there is a clear pattern in a scatter plot



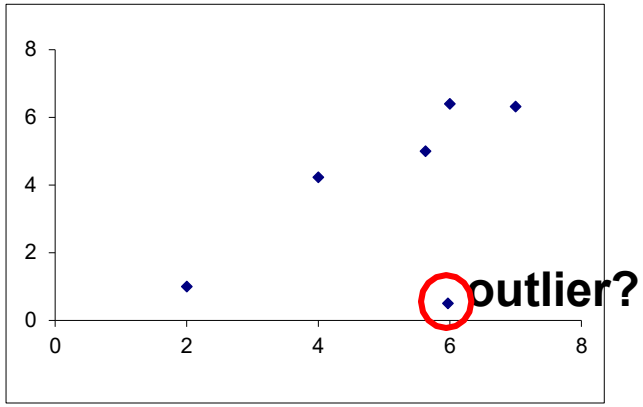
No clear pattern; less relationship.



Clear pattern: there is a **relationship**

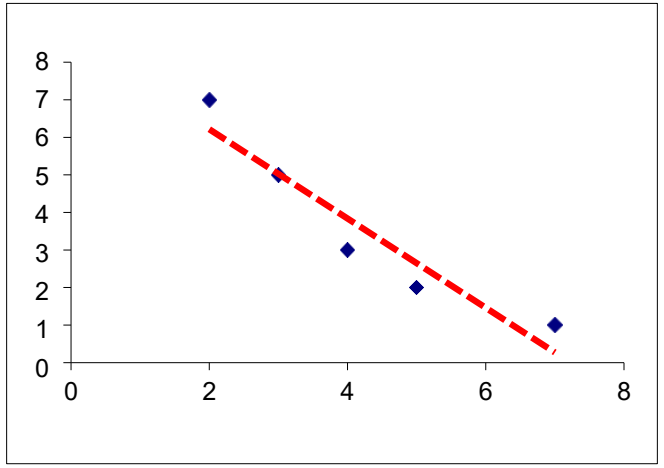


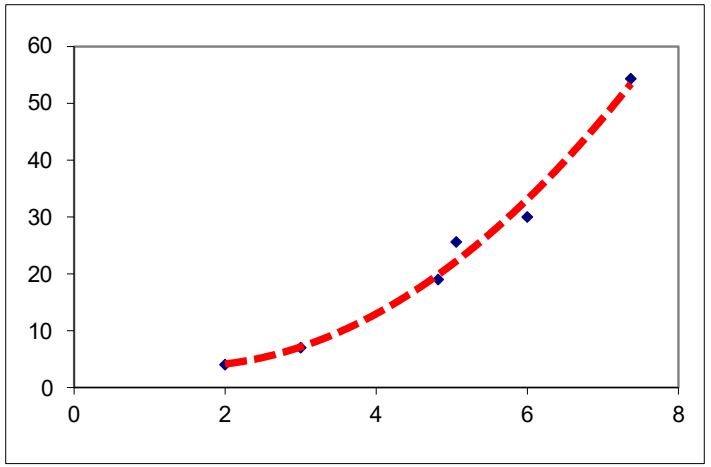
Points drift downwards; negative relationship; no outliers.



Points drift upwards; positive relationship; there seems to be one outlier.

How to Interpret a Scatter Plot…





- points has a linear form;
- points are scattered randomly around the straight line:
- linear relationship.

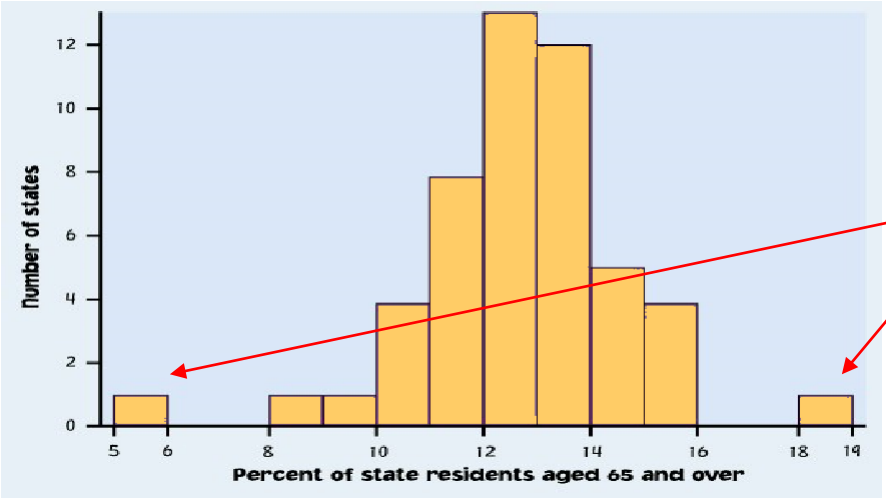
- points has a non-linear form;
- points are scattered randomly around the nonlinear curve
- non-linear relationship.

✓ Many non-liner relationships may be converted to linear by a transformation

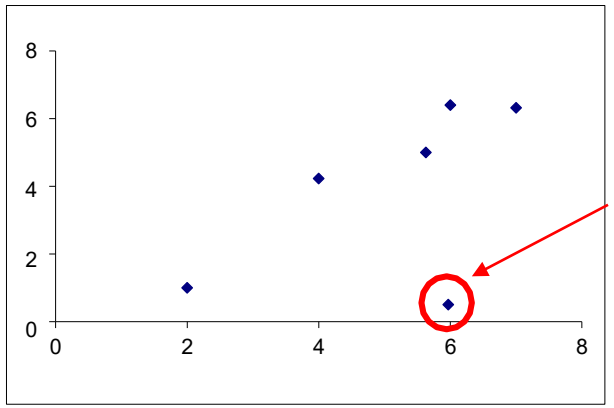
Outliers

Outliers

- An important kind of deviation is an outlier.
- Observations that lie outside the overall pattern
- Try to explain them.
- Large gap in the distribution is typically a sign of an outlier.



outlier?



outlier?

Principles of Excellent Graphs

- The graph should not distort the data.
- The graph should not contain unnecessary adornments (sometimes referred to as chart junk).
- The scale on the vertical axis should begin at zero.
- All axes should be properly labeled.
- The graph should contain a title.
- The simplest possible graph should be used for a given set of data.

Graphical Errors: Chart Junk

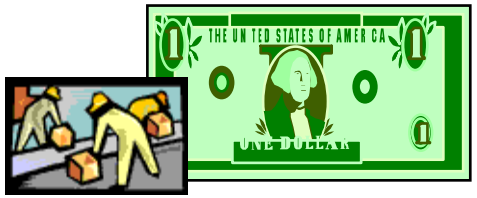
Bad Presentation

Minimum Wage

1960: $1.00

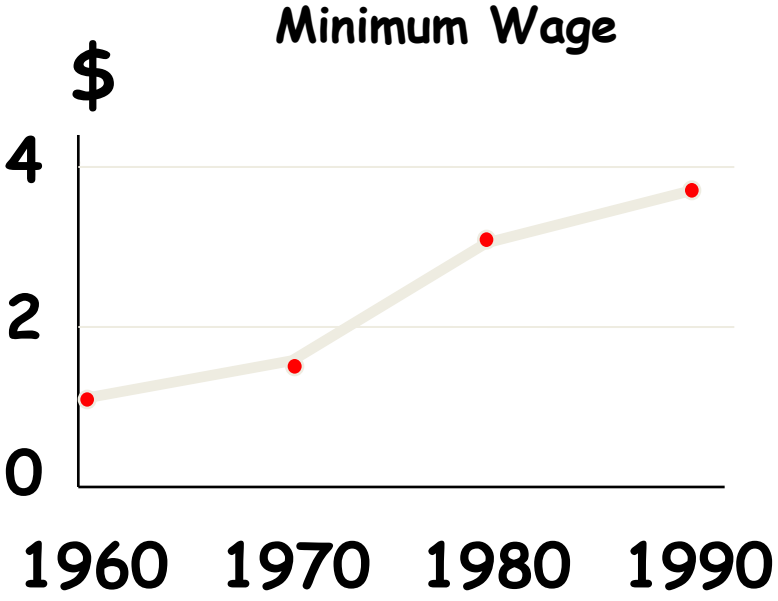
1970: $1.60

1980: $3.10



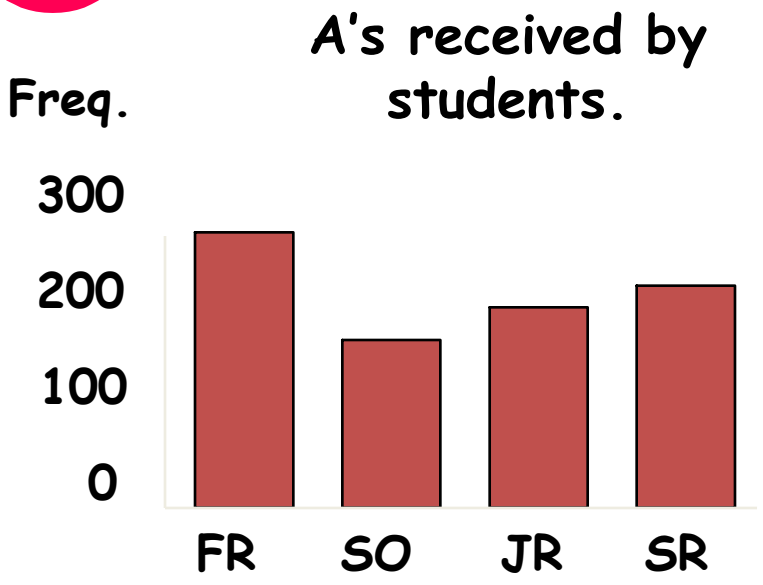
1990: $3.80

Good Presentation

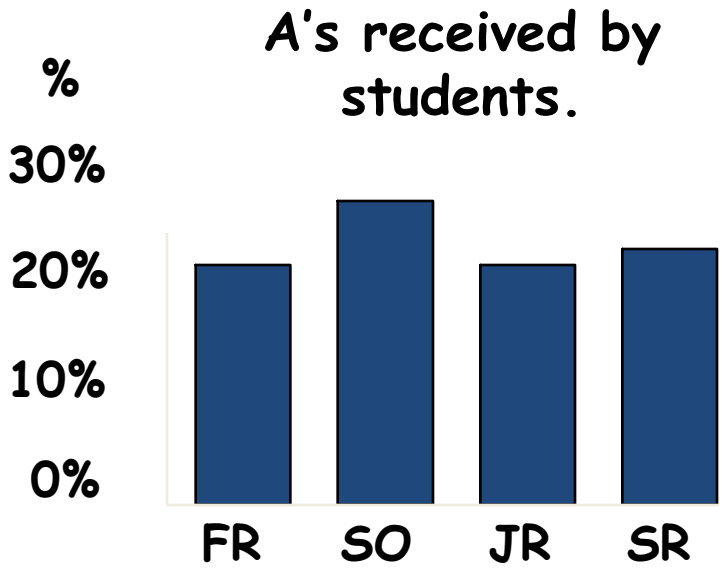


Graphical Errors: No Relative Basis

Bad Presentation



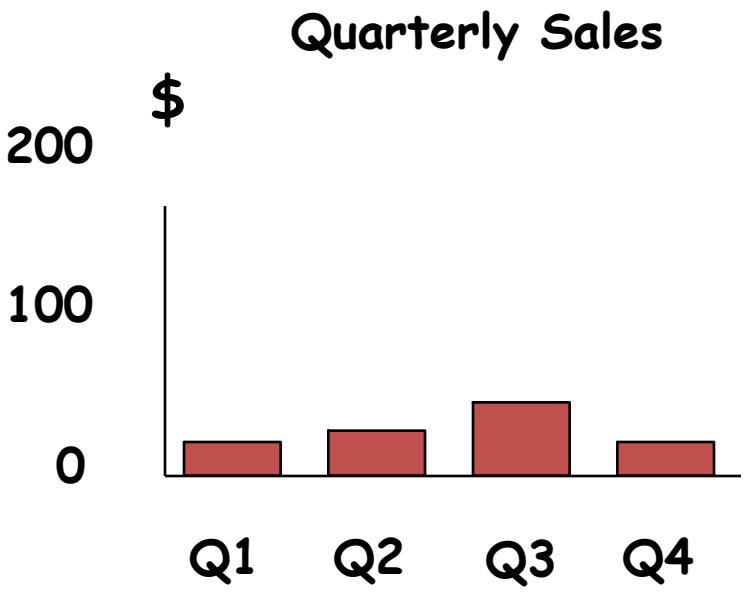
✓ Good Presentation



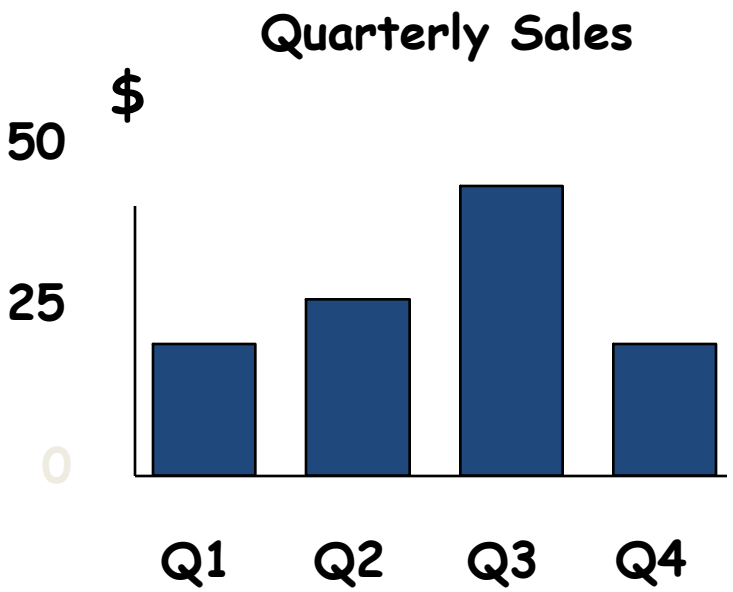
FR = Freshmen, SO = Sophomore, JR = Junior, SR = Senior

Graphical Errors: Compressing the Vertical Axis

Bad Presentation

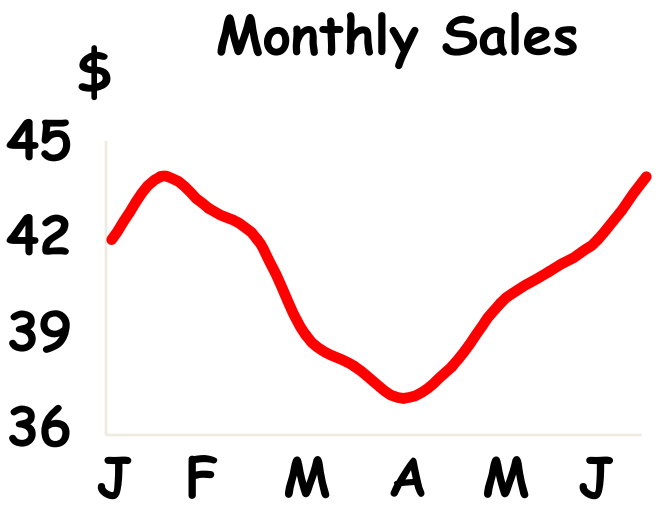


✓ Good Presentation

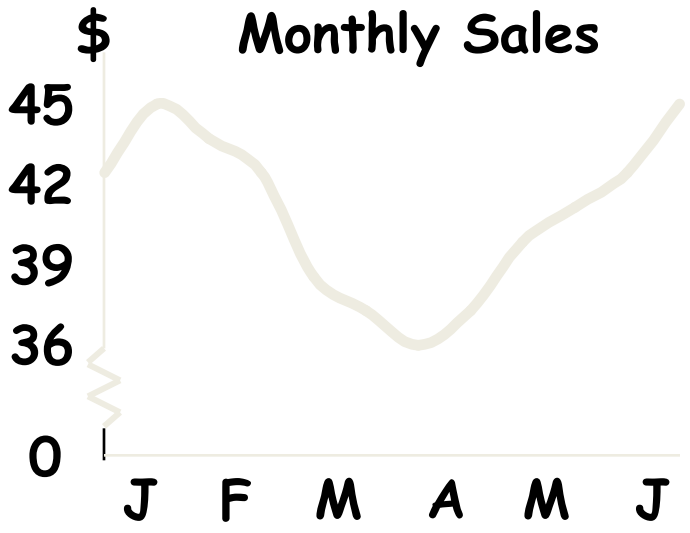


Graphical Errors: No Zero Point on the Vertical Axis

Bad Presentation



Good Presentations



Graphing the first six months of sales

Any questions?

Answers?

1. Explain separately the occasions that the bar charts, pareto charts and pi charts are used.
2. What sort of chart can be used to illustrate time series data?
3. Provide an example for the symmetric type distribution.
4. Describe what an outlier means.
5. Provide the answers for the question in the next slide.

A sample of 500 shoppers was selected in a large metropolitan area to determine various information concerning consumer behavior. Among the questions asked was "Do you enjoy shopping for clothing?" The results are summarized in the following table:

Enjoy shopping for clothing	Gender		
	Male	Female	Total
Yes	136	224	360
No	104	36	140
Total	240	260	500

a) Construct contingency tables based on total percentages, row percentages, and column percentages.

b) What conclusions do you reach from these analyses?